

Certifications for DevOps

Certifications can significantly enhance a DevOps career by validating one's skills and knowledge, demonstrating a commitment to professional growth, and often leading to better job opportunities and higher salaries. Here are some of the most prominent and respected certifications in the DevOps field.

AWS Certified DevOps Engineer – Professional

The AWS Certified DevOps Engineer – Professional certification focuses on provisioning, operating, and managing distributed application systems on the AWS platform. It is designed for individuals who perform a DevOps Engineer role with two or more years of experience provisioning, operating, and managing AWS environments. Key areas covered include:

- **Continuous delivery and automation:** Implementing and managing continuous delivery systems and methodologies on AWS.
- **Security controls:** Implementing and automating security controls, governance processes, and compliance validation.
- **Monitoring and logging:** Designing, managing, and maintaining tools to automate operational processes.
- **Incident and event response:** Automating monitoring, alerting, and auditing.
- **Infrastructure as code (IaC):** Using AWS CloudFormation and other tools to deploy infrastructure as code.

Certified Kubernetes Administrator (CKA)

The Certified Kubernetes Administrator (CKA) certification validates skills in Kubernetes, which is essential for managing containerised applications. This certification is aimed at Kubernetes administrators, cloud administrators, and other IT professionals responsible for Kubernetes management. Key areas covered include:

- **Cluster architecture, installation, and configuration:** Setting up and configuring a Kubernetes cluster, understanding the Kubernetes API, and deploying applications.
- **Workloads and scheduling:** Managing deployments, rolling updates, and scaling applications.
- **Services and networking:** Configuring network policies, service discovery, and load balancing.
- **Storage:** Provisioning and managing storage in Kubernetes.
- **Troubleshooting:** Diagnosing and fixing common issues in a Kubernetes environment.

Docker Certified Associate

The Docker Certified Associate (DCA) certification demonstrates expertise in Docker, an essential containerisation tool in DevOps. This certification is intended for professionals with some experience in Docker and containerisation. Key areas covered include:

- **Orchestration:** Understanding and using Docker Swarm and Kubernetes for orchestration.
- **Security:** Applying security best practices in Docker, including image scanning and vulnerability management.
- **Networking:** Configuring and troubleshooting Docker networking.
- **Storage and volumes:** Managing Docker storage and volumes.
- **Image creation and management:** Building, tagging, and pushing Docker images.

Microsoft Certified: DevOps Engineer Expert

The Microsoft Certified: DevOps Engineer Expert certification covers designing and implementing strategies for collaboration, code, infrastructure, source control, security, compliance, continuous integration, testing, delivery, monitoring, and feedback. It is ideal for professionals working with Microsoft Azure. Key areas covered include:

- **DevOps strategy:** Developing an instrumentation strategy, site reliability engineering (SRE) strategy, and security and compliance plan.
- **Implementation:** Implementing continuous integration, delivery, and feedback.
- **Infrastructure as code (IaC):** Managing configuration and infrastructure as code using tools like Azure Resource Manager, Terraform, and Azure DevOps.
- **Dependency management:** Managing dependencies and package management using tools such as NuGet, npm, and Maven.
- **Application infrastructure:** Implementing application infrastructure by using Azure Kubernetes Service (AKS) and other Azure services.

Additional certifications

While the above certifications are among the most recognised, there are other valuable certifications that can enhance a DevOps career.

Google Professional Cloud DevOps Engineer: Focuses on managing service availability and delivery using Google Cloud.

Red Hat Certified Specialist in OpenShift Administration: Demonstrates skills in managing Red Hat OpenShift Container Platform environments.

HashiCorp Certified: Terraform Associate: Validates proficiency in infrastructure as code using Terraform.

A few ways of gaining experience

Practical experience is critical in a DevOps career. Working on real-world projects, either through internships, co-ops, or entry-level positions, helps build the necessary skills and provides valuable insights into the practical aspects of DevOps. Internships and co-ops offer a structured